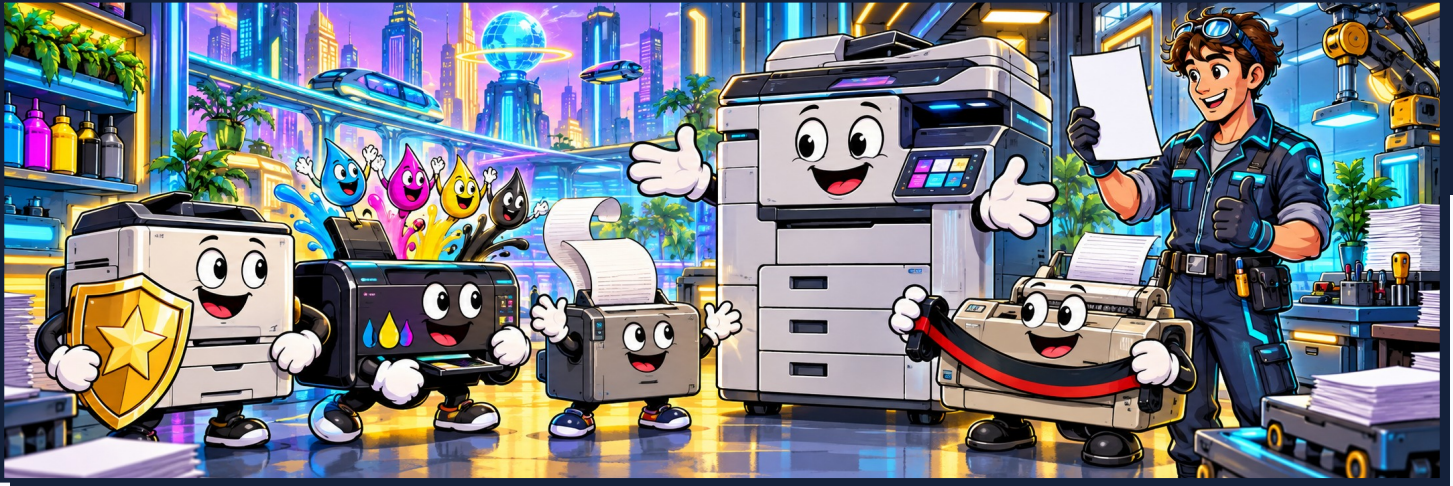


NETWORK CITY

PRINTERS & SCANNERS

Print, maintain, and troubleshoot



LEARN THE PARTS. FOLLOW THE PAGE. SOLVE THE SCENARIO.

Detailed comic illustrations + short definitions + clear exam clues

Explore laser, inkjet, thermal, and impact printers. Learn setup, maintenance, and troubleshooting for objectives 3.7, 3.8, and 5.6. Practice 24 flashcards in the matching clickable PowerPoint.

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- 03 Speed, detail & workload words
- 04 Four printer families
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- 06 The seven-step laser memory map
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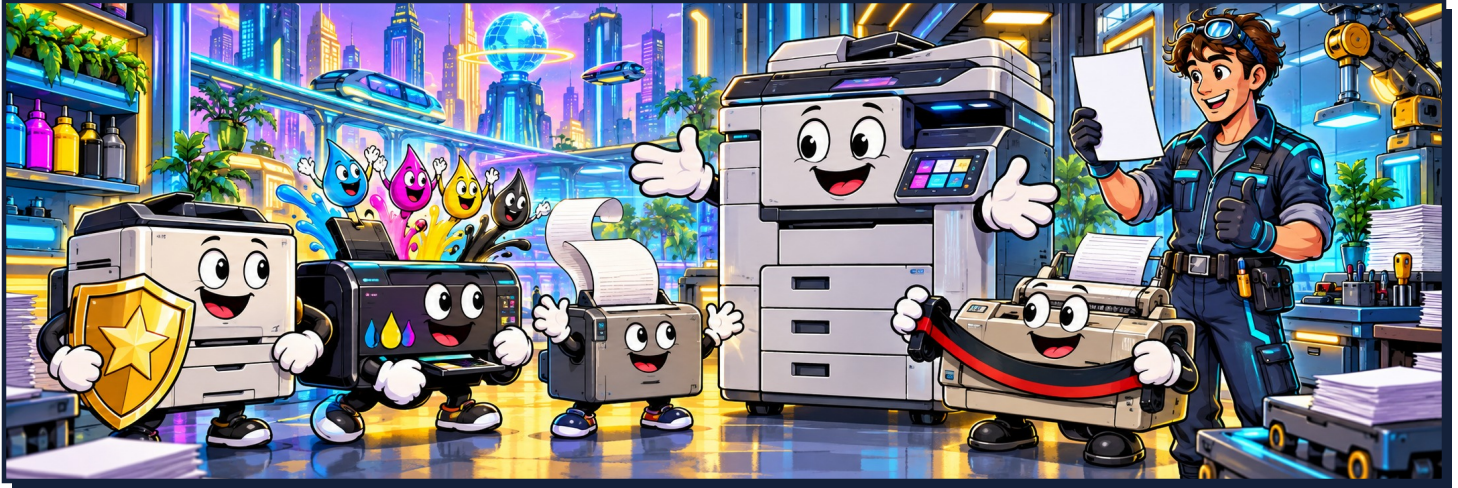
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Independent study guide. Aligned to exam 220-1201; not an official CompTIA publication. Illustrations are memory aids. Technical notes and labeled diagrams explain the real rules.

01 / PRINTERS AND SCANNERS

Meet the printer crew

●●● **VISUAL:** Draw googly-eyed laser, inkjet, thermal, and impact heroes in a neon cyan and purple print shop.



WHIRR! We turn digital work into paper!

WHIRR!

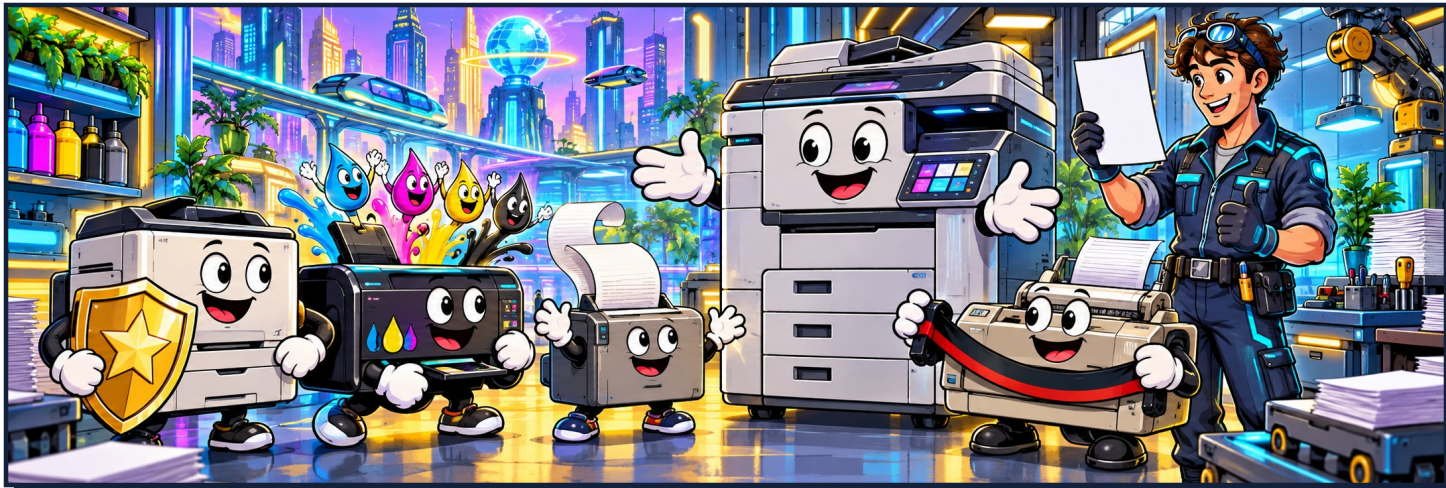
- **Printer.** An output device. It puts text or pictures onto paper or another supported material. A print job is the document plus instructions, such as copies and paper size.
- **MFP - Multifunction Printer.** One machine combines printing with scanning and copying. Some models also fax. MFD means Multifunction Device. Check the actual features before choosing one.

- **Print, scan, copy.** Print makes a physical page from digital data. Scan turns a physical original into a digital image. Copy scans an original and prints a duplicate.
- **Consumable.** A supply that runs out through use, such as toner, ink, ribbon, or paper. A wear part, such as a roller, needs service after use too.
- **The team rule.** Match the printer, supplies, paper, driver, and settings. A fast engine cannot fix the wrong paper or a job sent to another printer.

EXAM MAP: A+ Core 1, 220-1201. Objectives 3.7: setup and settings; 3.8: maintenance; 5.6: troubleshooting. The laser cycle adds useful background for understanding the parts. [1]

Speed, detail & workload words

VISUAL: Draw a neon printer hero with a stopwatch, a magnifying glass, and a tidy stack of paper.



Fast, sharp, and busy mean different things!

ZOOM!

• **PPM - Pages Per Minute.** A print-speed measure. Color, paper size, quality, warm-up, and two-sided printing affect actual speed. Compare ratings made under the same test conditions.

• **DPI - Dots Per Inch.** The density of printed dots. At 600 DPI, there can be 600 dot positions along one inch. DPI is about detail; it does not mean 600 pages each minute.

• **First-page-out time.** How long the first finished sheet takes under stated conditions. A printer can start slowly yet print a long job quickly once it warms up.

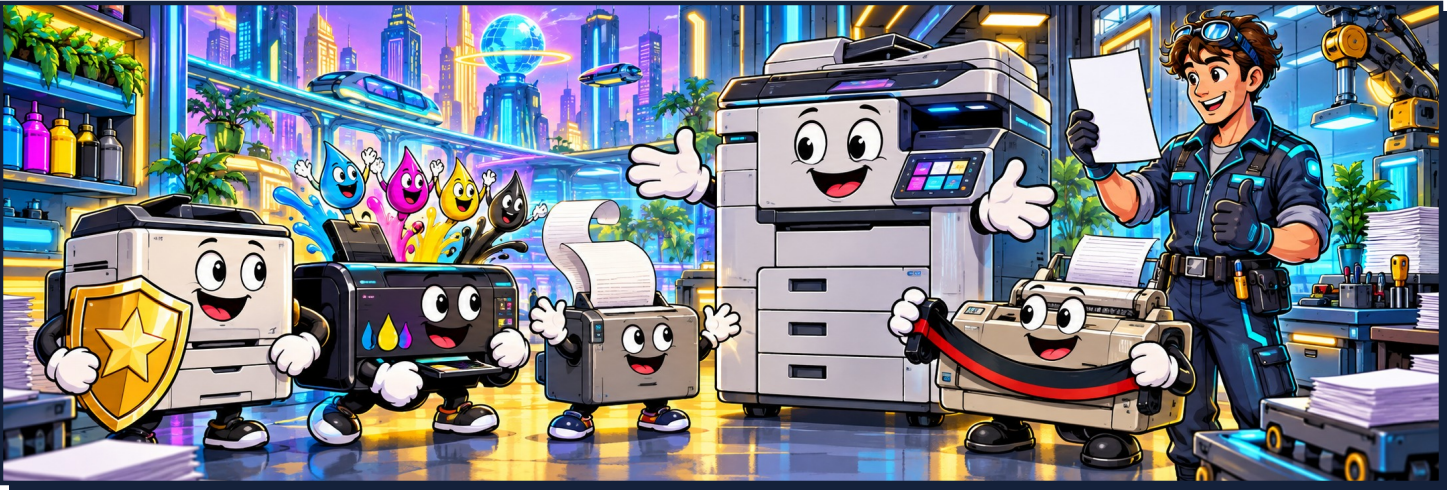
• **Duty cycle.** A stated maximum monthly workload. The recommended monthly page volume describes the normal workload the maker suggests. Do not treat the maximum as a routine target. [18]

• **Yield / cost per page.** Yield is estimated pages from a supply under a test method. Heavy ink or toner coverage lowers actual yield. Compare supply cost divided by expected pages, plus other running costs.

MEMORY HOOK: PPM = pace. DPI = detail. Recommended volume = routine workload. No single speed applies to all laser or inkjet printers.

Four printer families

●●● **VISUAL:** Draw four bright printer characters lining up for office, photo, receipt, and multipart-form jobs.



Pick the team that fits the mission!

PICK!

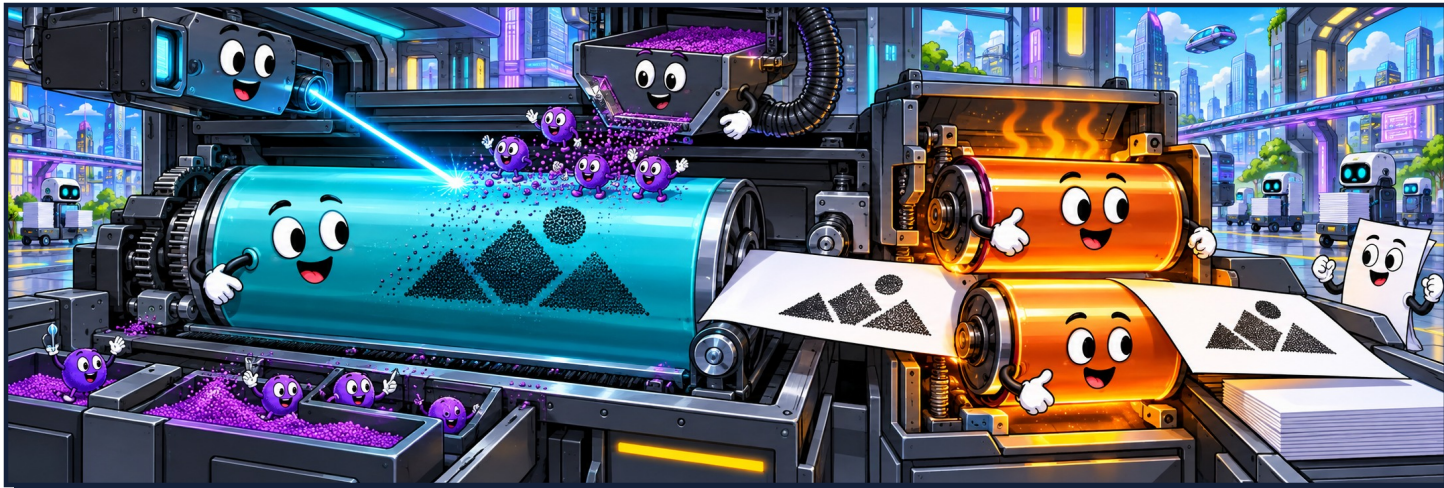
Type	How it makes marks	Common use
Laser	Toner powder, charge, drum, and fuser	Office documents; repeated text jobs
Inkjet	Tiny liquid ink drops from nozzles	Color work and photos on suitable paper
Direct thermal	Heat changes special coated paper	Receipts and short-life labels
Impact / dot matrix	Pins strike an ink ribbon	Multipart forms and legacy records

- **A useful distinction.** Laser and inkjet are non-impact methods: they do not stamp a ribbon into a stack of forms. An impact printer uses a physical strike. [2,5,7]
- **Thermal transfer.** Another thermal method uses heat and a ribbon to mark labels. Learn it beside direct thermal so you choose the right supply. [5]

CHECK THE JOB: Photos suggest a suitable inkjet. Multipart forms suggest impact. A receipt roll may suggest direct thermal. Actual product support decides the final choice.

Laser parts: the powder factory

VISUAL: Draw a blue drum, purple toner sprites, and gold fuser rollers inside a bright comic factory.



Powder first. Heat and pressure later!

ZAP!

- **Toner.** Fine powder that includes color pigment and a plastic-like binder. It is not liquid ink. The fuser bonds it to paper. Monochrome printers usually use black toner.

- **Imaging drum / photoconductor.** A light-sensitive surface that holds an electrical charge pattern. Think of a reusable image map. Avoid touching its surface or leaving it in bright light.

- **Charge roller / corona wire.** Applies an even starting charge to the drum. Designs differ. A laser then changes the charge in selected places to write the image.

- **Developer system.** Presents charged toner to the image areas. A transfer roller or belt helps move the toner toward the paper.

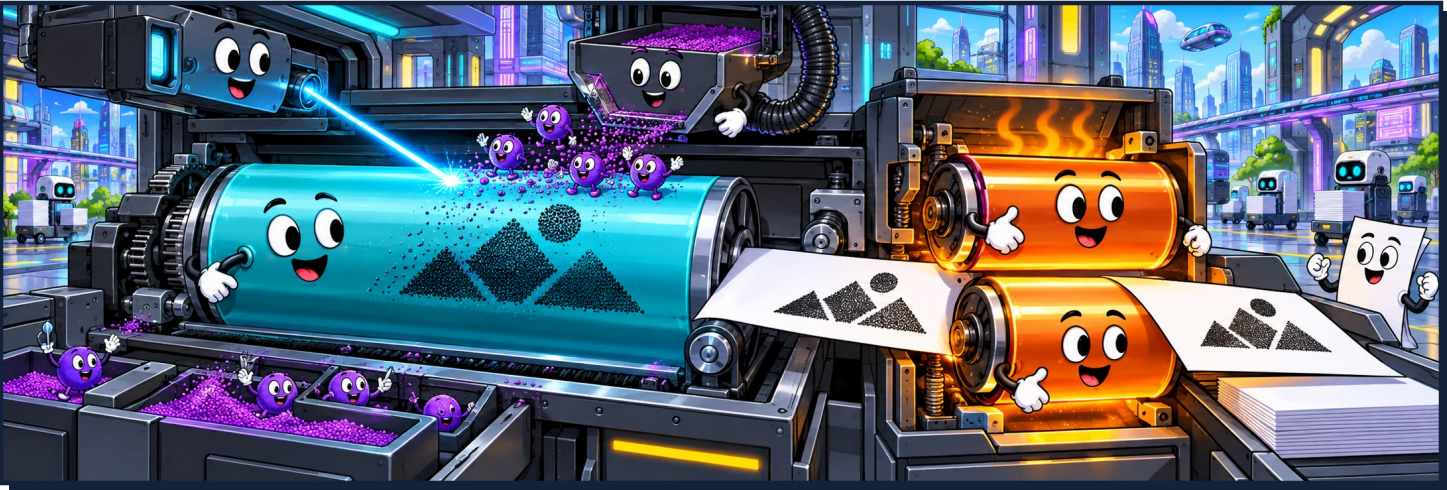
- **Fuser.** Uses heat and pressure to bond toner to the page. It is a different part from the drum and can stay hot after printing. [2,3]

- **CMYK.** Cyan, Magenta, Yellow, and Key (black). These are common process-printing colors. Color lasers combine toner images, often through a transfer belt.

PARTS VARY: Some cartridges combine toner and drum. Other machines use separate units. Read the model manual before ordering or replacing a part.

The seven-step laser memory map

VISUAL: Draw the neon toner factory with its drum and fuser. Put the exact sequence in the clear table below.



Build the image, move it, and make it stay!

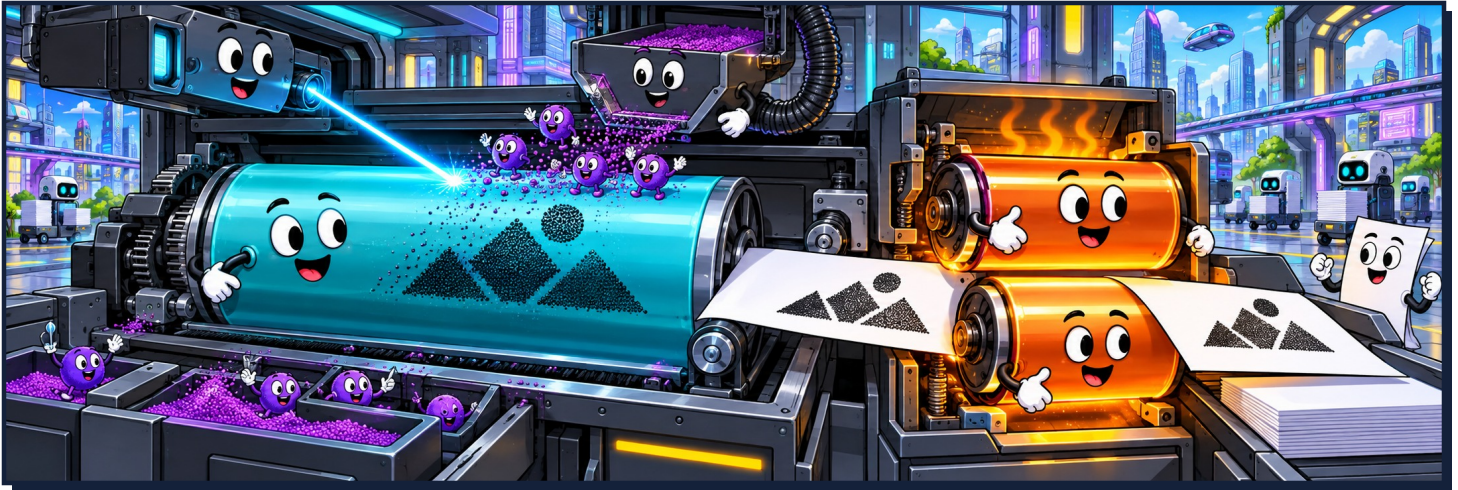
WHIRR!

Order	Stage	Main job
1	Processing	Make the page image from job data
2	Charging	Give the drum an even starting charge
3	Exposing	Light writes the charge pattern
4	Developing	Toner forms the image on the drum
5	Transferring	Move the toner image toward the paper
6	Fusing	Bond toner with heat and pressure
7	Cleaning	Remove leftover toner for reuse of the drum

SAY IT: Processing, Charging, Exposing, Developing, Transferring, Fusing, Cleaning. Memory sentence: “Please Charge Every Drum Then Fuse Cleanly.” Real machines cycle continuously. [2,3]

Laser steps 1-3: write the image

VISUAL: Draw a friendly processor handing a page map to a blue drum inside the cyan-lit laser factory.



ZAP! Light writes a hidden picture!

ZAP!

- **1. Processing.** The computer and printer turn the print job into an image the engine can make. A raster image is a grid of tiny dots. RAM means Random Access Memory; it temporarily holds working data.
- **2. Charging.** A charge roller or corona wire gives the drum an even electrical charge. The drum starts with a clean electrical backdrop.
- **3. Exposing.** A laser changes the charge in selected places on the drum. This creates a latent image: a hidden charge pattern that will guide toner. Some similar engines use LEDs, Light-Emitting Diodes.

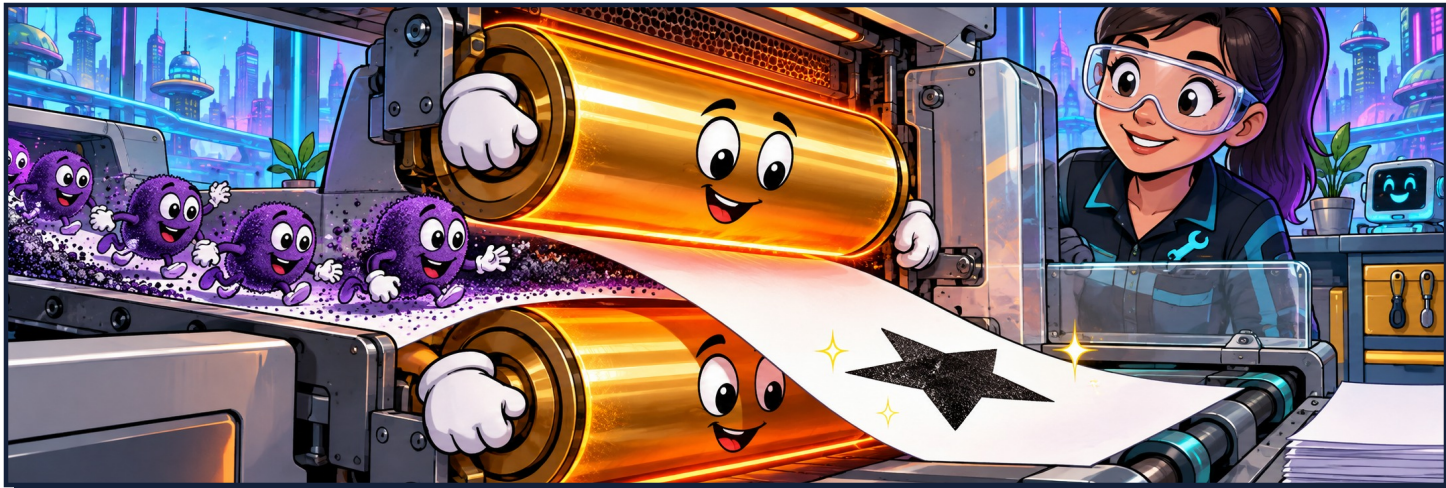
• **Why charge matters.** Electrical forces control where toner goes. The laser writes on the drum; it does not burn letters directly into the paper. Exact voltages and polarities depend on the design.

• **Use this clue.** Bad job data or an unsupported printer language can cause garbled output even when the mechanical parts work. Separate a data problem from an engine problem. [2,8]

RECALL CHECK: Which part receives the light pattern? The drum. Which stage creates the dot map? Processing.

Laser steps 4-7: finish the page

VISUAL: Draw gold fuser rollers pressing a toner picture into paper while a technician watches behind a guard.



PRESS! Now the image stays on the sheet!

PRESS!

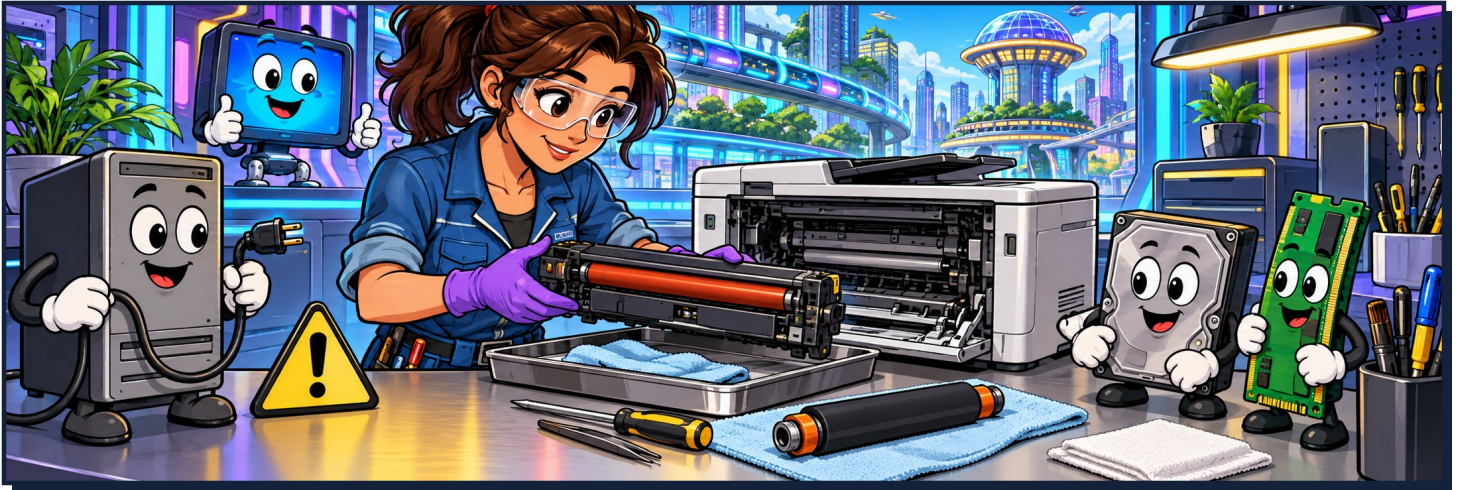
- **4. Developing.** The developer system brings charged toner to the drum. The charge pattern controls which areas receive toner, making the hidden image visible.
- **5. Transferring.** Electrical forces move the toner image from the drum toward the paper. Some color designs use an intermediate transfer belt first.
- **6. Fusing.** Heat softens the toner binder. Pressure helps bond it to the paper. Before fusing, loose toner can rub off. A fuser is not an ink-drying fan.

- **7. Cleaning.** A blade or other cleaning part removes toner left on the drum. The printer prepares the drum for the next image. Waste toner goes to a collection area.
- **Continuous cycle.** Different areas of the rotating drum can be at different stages at the same time. The seven-step list is a learning sequence, not seven separate pauses. [2,3]

HOT CLUE: Toner that rubs off can point to poor fusing. Check the paper type and settings too before replacing the fuser. [14]

Laser maintenance: keep it clean

VISUAL: Draw a careful tech, a cooled service part, and a disconnected plug on a purple and cyan workbench.



Check the manual before the swap!

CHECK!

- **Toner replacement.** Use the correct cartridge. Follow its handling and seal-removal steps. Keep the drum protected. Install until seated, close the covers, and print a test page.

- **Maintenance kit.** A model-specific group of wear parts. It may include a fuser, pickup rollers, separation parts, or transfer parts. Kit contents and replacement counts vary. [3]

- **Calibration.** The printer adjusts its output to a target. Color calibration may improve density or color alignment. Use the built-in procedure and supported supplies.

- **Cleaning.** Use the maker's cleaning page or approved cloth and method. A cleaning page runs with power on. Physical internal service may require shutdown, unplugging, and cooling. Follow the task instructions. [13]

- **After service.** Test output and feeding. Reset a maintenance counter only when the required work is complete and the manual directs it. Record the part and page count.

SUPPLY CLUE: "Replace toner" and "replace drum" may describe different jobs. Do not assume every cartridge or maintenance kit includes both.

Inkjet: the tiny-drop artists

VISUAL: Draw cyan, magenta, yellow, and black ink characters helping a googly-eyed printhead paint a page.



SPLAT! Tiny drops build the picture!

SPLAT!

- **Inkjet.** A printer that places tiny liquid ink drops on paper through nozzles. Many heads move across the sheet; some designs use a fixed print bar.
- **Printhead / nozzle.** The head holds the tiny openings that aim the drops. Dried ink or debris can block a nozzle and leave gaps or bands.
- **Cartridge / tank.** A cartridge is a replaceable ink container. A tank system stores refill ink. A cartridge may include a printhead, but many printers keep the head separate.

- **Thermal / piezoelectric.** Two ways to push ink out. Thermal inkjet uses tiny heat-made bubbles. Piezoelectric inkjet uses a material that changes shape with an electrical signal. This differs from direct thermal paper printing.
- **Use cases.** Photos, colorful projects, and mixed home or office jobs. Paper coating and the selected media setting affect how ink spreads and dries.
- **Feed parts.** Rollers move sheets. A feeder supplies them. The paper path must stay clear even when the ink system works perfectly.

MEMORY HOOK: Inkjet = liquid drops and nozzles. Laser = powder toner and a fuser. Use the correct ink and paper for the model. [4,9,24]

Inkjet care: check, clean, align

VISUAL: Draw the tech comparing a gapped nozzle pattern and neat lines under large magnifying glasses.



Test the pattern before you chase the problem!

TEST!

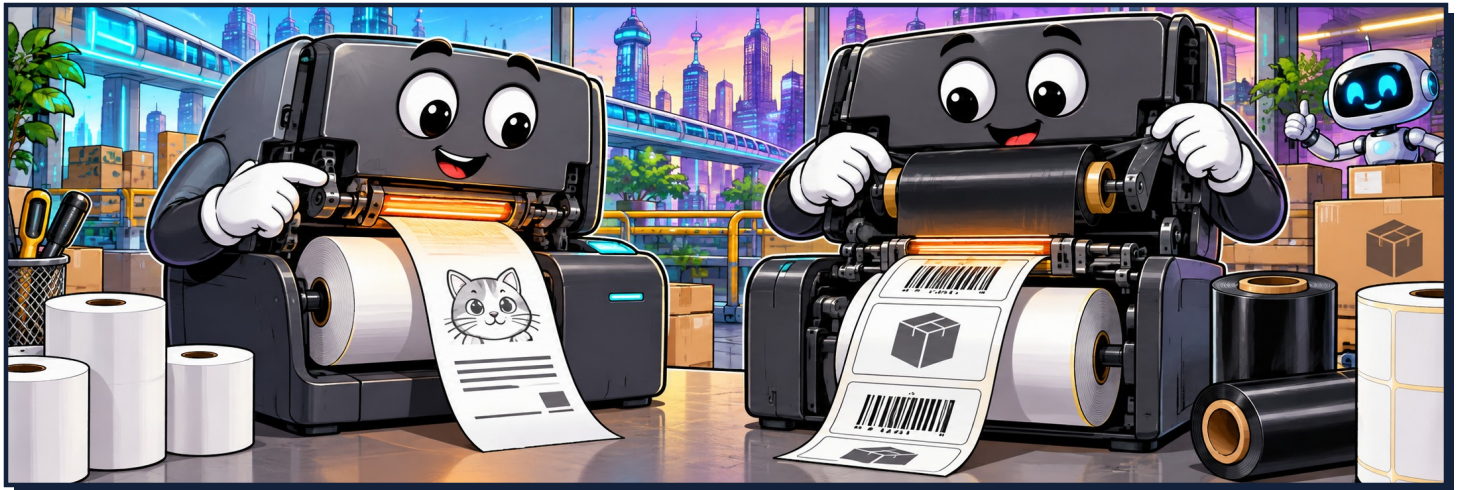
- **Nozzle check.** Prints a pattern that reveals missing jets. Gaps suggest a nozzle or ink-supply problem. Check ink level and correct cartridge installation too.
- **Head cleaning.** Runs a maintenance routine to clear nozzles. It uses ink. Follow the manual's retry limit and wait time instead of endlessly repeating cleaning. [4]
- **Alignment.** Adjusts where ink lands. It helps with doubled, offset, or jagged lines. Clean blocked nozzles first so the alignment pattern can be read correctly. [9]

- **Replace or refill.** Use the specified ink. Remove only the tape or seals the instructions identify. Do not touch nozzle plates or electrical contacts. Follow the model's setup routine afterward.
- **Rollers and jams.** Load supported paper squarely. Follow the approved roller-cleaning method. Keep liquid away from electronics and avoid forcing a stuck carriage.

SCENARIO: Missing bars in a nozzle test? Check ink and clean as directed. Complete bars that print out of line? Consider alignment. These are different maintenance jobs.

Thermal: the heat-printing twins

VISUAL: Draw two friendly neon receipt printers. One has only coated paper; the other has a separate dark ribbon.



Same heat idea. Different supplies!

WARM!

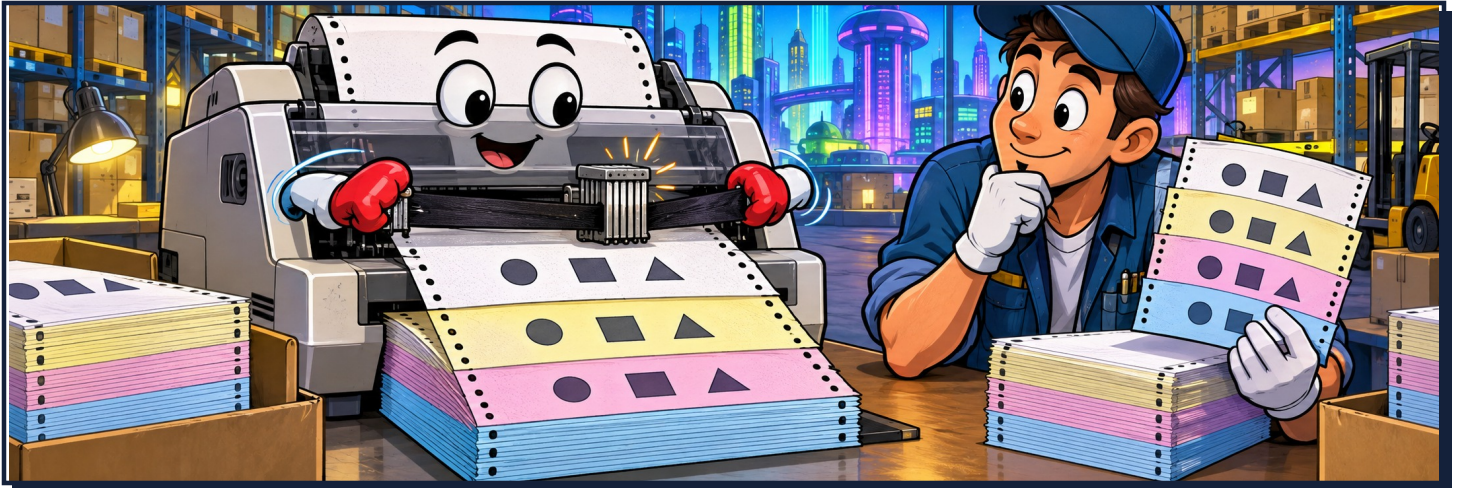
- **Direct thermal.** A heated printhead darkens heat-sensitive paper. It needs no ink, toner, or ribbon. Common uses include receipts and short-life shipping labels. [5]
- **Thermal transfer.** A heated head transfers material from a ribbon onto the label. The ribbon and label must work together. Suitable combinations can make durable labels. [5]

- **Feed assembly / platen.** The feed parts move media. A platen roller supports it at the printhead. Dust or adhesive buildup can affect feeding and print quality.
- **Care.** Load the right media with the printable side toward the head. Let the head cool. Use the model-approved cleaning pen, swab, or solution. Remove debris without scratching the surface. [6]
- **Clues.** Blank direct-thermal output can mean the wrong paper or wrong side facing the head. A line that stays blank may involve dirt or a damaged head element. Test before replacing parts.

MEMORY HOOK: Direct = special paper. Transfer = ribbon plus suitable media. Keep direct-thermal records away from excess heat and sunlight, which can damage readability.

Impact: the ribbon punch team

●●● **VISUAL:** Draw a tough dot-matrix hero with boxing gloves, an ink ribbon, and fanfold forms in a neon warehouse.



TAP-TAP! One strike can mark several layers!

TAP!

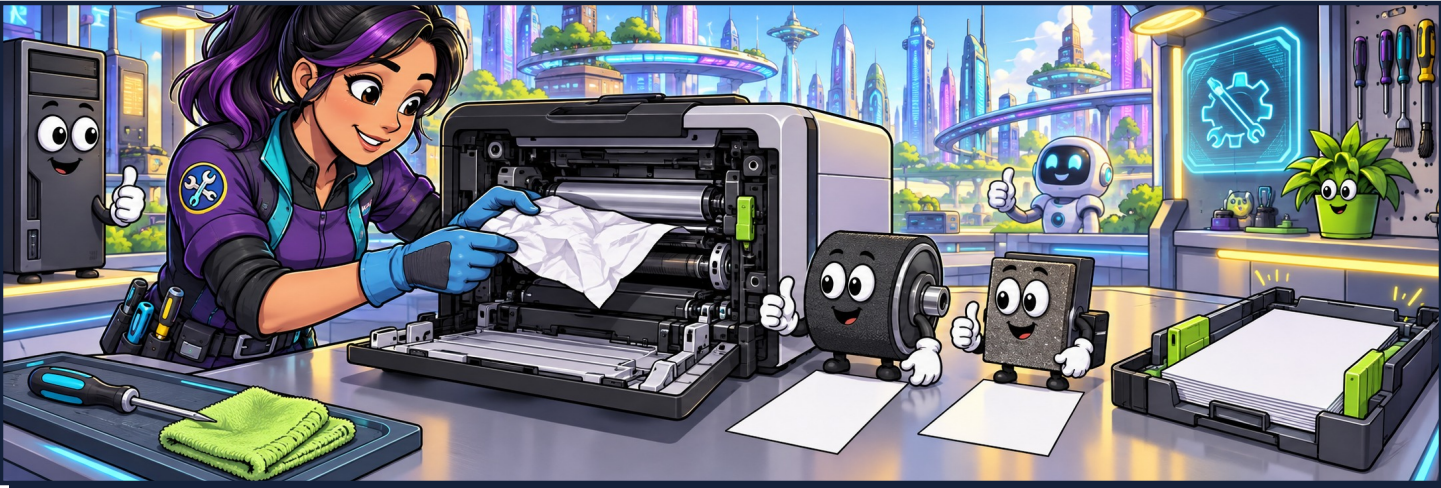
- **Impact / dot matrix.** Pins strike an ink-soaked ribbon against paper. The dots form letters and images. The physical strike can make marks through supported multipart forms.
- **Multipart forms.** Stacked forms that make copies from one printing pass. Carbonless forms use pressure-sensitive layers. The maximum number of parts depends on the printer. [7]
- **Tractor feed.** Sprockets grip holes along the sides of continuous paper. They help pull fanfold forms through in line. Single-sheet feeding may also be supported.

- **Ribbon / printhead.** Replace a worn ribbon for weak printing when appropriate. Missing dots may point to a head or pin problem. Let the head cool before service.
- **Head gap / paper thickness.** The spacing must suit the form thickness. Too little or too much space can cause jams or poor marks. Use the manual's setting rather than guessing.
- **Care.** Load straight, match the form size, and keep printing inside the allowed area. Replace the ribbon, head, and paper using the correct model procedure.

EXAM CLUE: A noisy printer that uses a ribbon and multipart tractor-feed forms strongly suggests impact printing. [1,7]

Paper: inches, coating & storage

VISUAL: Draw cheerful Letter and Legal sheets beside snug green paper guides in a colorful supply room.



The right sheet keeps the mission moving!

FIT!

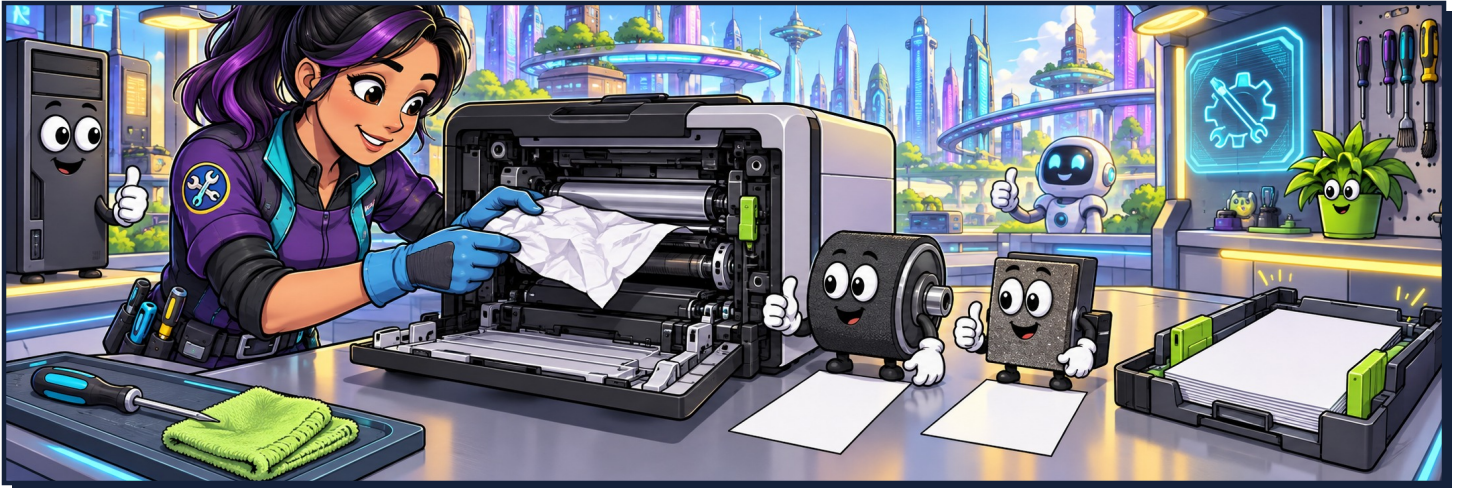
Paper name	Dimensions in inches	Common use
U.S. Letter	8.5 x 11	Everyday U.S. documents
U.S. Legal	8.5 x 14	Longer forms and legal documents
Tabloid / Ledger	11 x 17	Larger layouts; supported models only
A4	About 8.27 x 11.69	Common international office size

- Media.** The material the printer marks: paper, envelopes, cards, or labels. Match size, weight, coating, and heat tolerance to the printer and tray. [19]
- Storage and loading.** Keep paper dry, flat, and protected. Remove damaged sheets. Stay below the fill line. Set guides snugly without bowing the stack.

SIZE TRAP: Letter and A4 are different sizes. Match the app, driver, tray, and loaded sheet. Use only labels or transparencies approved for the printer’s heat and feed path.

Paper path: one sheet at a time

VISUAL: Draw a smiling pickup roller and a firm separation-pad guard guiding single sheets through a neon printer.



One sheet, please! Keep the line moving!

ROLL!

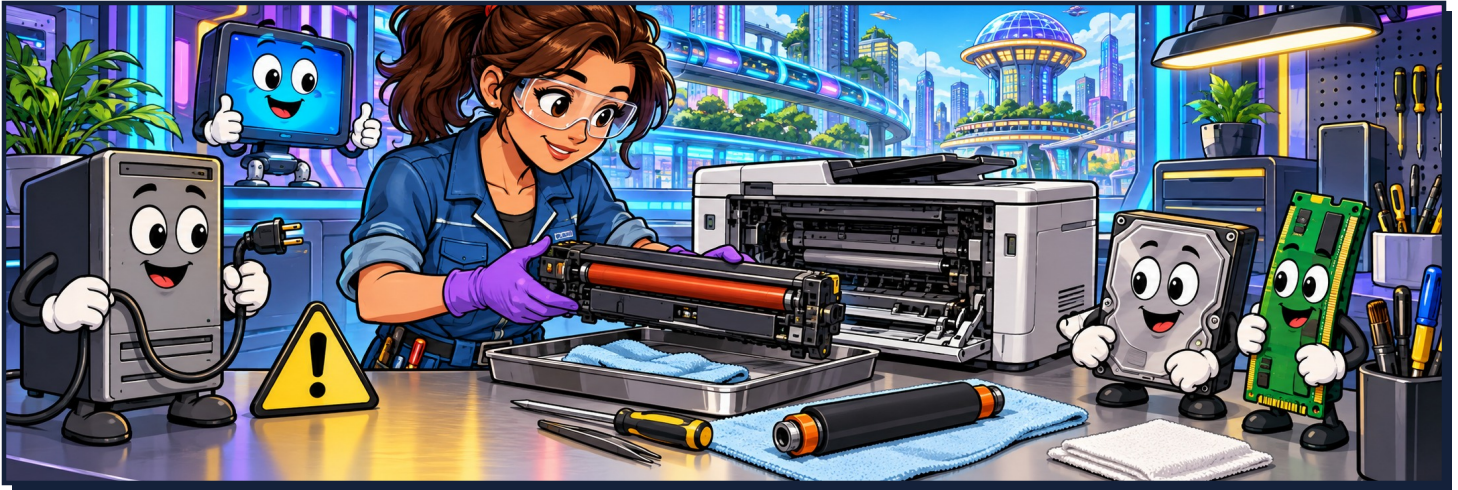
- **Pickup roller.** Grabs a sheet from the input stack. A dirty or worn roller may slip, leaving paper in the tray.
- **Separation pad / roller.** Helps hold back extra sheets. A worn separation part can allow two or more sheets to feed together. Paper condition matters too.
- **Registration rollers.** Help line up the sheet and time its arrival at the print area. Skew means a sheet sits or moves at an angle.

- **Duplexer.** Routes a printed sheet so the engine can print the other side. Automatic duplexing requires supported hardware and paper. Manual duplexing asks the user to reload sheets.
- **Finisher.** An optional unit that staples, punches holes, sorts, or folds. It has its own limits and can jam even when the printer engine works.

CLUE MAP: No pickup? Inspect paper and pickup parts. Several sheets? Check separation and the stack. Back-side jam? Check the duplex path. [14,20]

Unbox & choose a safe location

VISUAL: Draw a technician checking a sturdy level table, unpacking parts, and keeping the neon printer's vents clear.



A good setup prevents the first jam!

READY!

- **Unbox correctly.** Use the lift points and enough help for the weight. Remove shipping locks, tape, and packing pieces the instructions identify. Keep the packaging list handy.

- **Location.** Choose a stable surface that supports the printer. Leave the maker's clearance around vents, trays, and service doors. Keep food, drinks, heat, and damp areas away.

- **Power.** Match the voltage and power requirements. Use the proper grounded connection where required. Follow the maker's guidance for power strips or backup power. Laser warm-up can draw substantial power.

- **Safety signs.** Hot-surface labels mean wait for cooling before touching service parts. Keep hands, hair, and jewelry away from moving mechanisms. Do not bypass door interlocks.

- **Physical service.** Shut down, unplug, and cool the unit when the service procedure requires it. Keep power supplies and enclosed laser assemblies closed unless trained and authorized. [12]

ESD = Electrostatic Discharge: a static zap can harm electronics. Use the approved ESD procedure for exposed circuit boards. Never wear a grounding strap while working on energized power circuits.

Drivers, firmware & page languages

VISUAL: Draw a cyan laptop translator handing a clear page plan to a smiling printer in the neon office.



The computer and printer must speak the same language!

TALK!

- **Driver.** Software that helps the operating system send suitable print data and settings. OS means Operating System. Match the printer model, OS, and supported computer architecture.
- **PCL.** Printer Command Language, HP's printer-language family. The CompTIA setup objective also spells it "Printer Control Language." Recognize both wordings. PCL is common for office printing. [1,8,21]

- **PS - PostScript.** A page-description language associated with Adobe. It is useful in supported graphics workflows. The printer must support the selected language or an appropriate emulation.
- **Firmware.** Software stored in the printer that runs the device. Use the correct official update and a stable power connection. Do not interrupt an update. [22]
- **Driverless printing.** Compatible standards can let the OS print without a separate vendor driver package. Feature support still matters. A generic print path may not expose every scanner or finisher option. [10,11]

GARBLED OUTPUT? Check the selected printer, correct driver, language support, and connection. "Universal" does not mean every printer understands every language.

Connections: USB, Ethernet & Wi-Fi

VISUAL: Draw a blue Ethernet cable joining a printer and switch while another printer connects by USB.



Follow the path from your desk to the printer!

LINK!

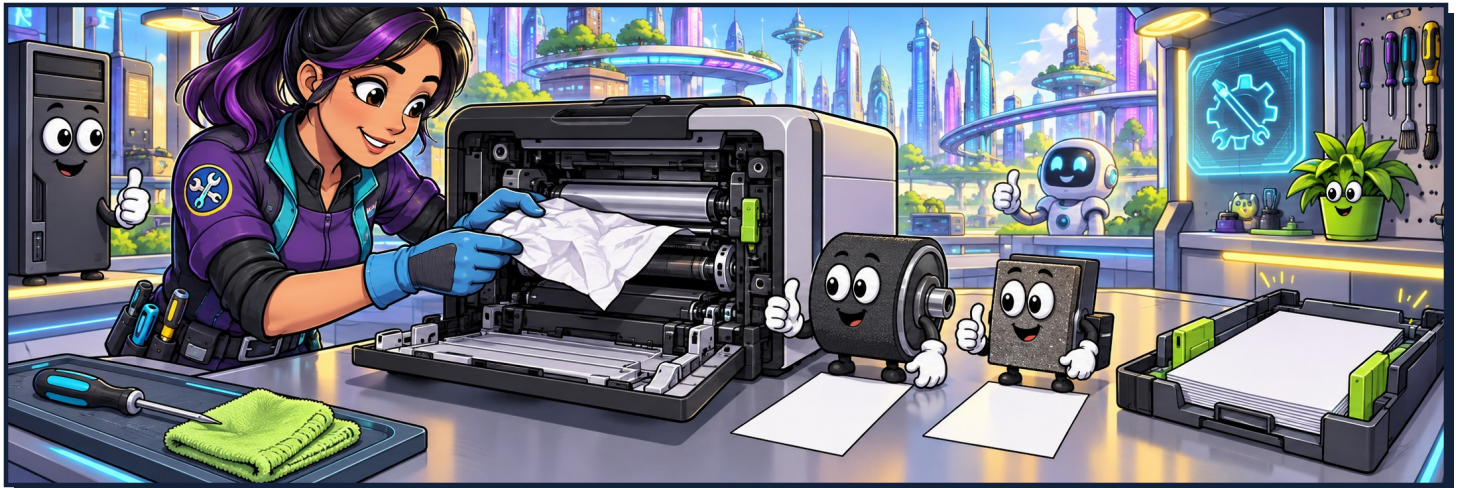
- **USB - Universal Serial Bus.** Often connects one computer directly to the printer. Use a data-capable cable and follow setup instructions about when to plug it in. A local connection can help isolate a network fault.
- **Ethernet.** A wired network connection. A network printer can serve several clients without using one person's computer as the host. Check cable seating and link indicators.
- **Wi-Fi.** Wireless local networking. Check the supported band, network name, credentials, and signal. A guest network may isolate clients and block local printing.

- **IP / DHCP / DNS.** Internet Protocol provides addresses. Dynamic Host Configuration Protocol assigns network settings. Domain Name System maps names to addresses. A changed address can break a queue using an old address.
- **Stable addressing.** An approved DHCP reservation keeps a device's assigned address consistent. A carefully configured static address is another option. Avoid duplicate addresses.

REACH CHECK: Confirm the printer's current address from its status page. Test access from the client. A reply to ping does not prove the print service works. [16]

Settings: both sides & the right tray

VISUAL: Draw a neat paper team with two-sided pages and snug guides beside a neon printer's trays.



Choose the page before pressing Print!

FLIP!

- **Simplex / duplex.** Simplex prints one side. Duplex prints both. Choose long-edge or short-edge binding to control which way the back side turns. Test one sheet before a large job.

- **Orientation.** Portrait stands taller than it is wide. Landscape sits wider than it is tall. Rotation, page size, and scaling can change how a document fits.

- **Tray settings.** Tell the printer and driver which size and media type each tray holds. A requested size or type that is unavailable can make a job wait.

- **Quality.** Draft or economy settings favor speed or lower supply use. Higher quality can take longer. Color mode, resolution, and media settings must suit the task.

- **Installed options.** The driver may need to detect or enable an added tray, duplexer, or finisher. Choosing an option in software cannot add missing hardware.

THREE-WAY MATCH: Check the document settings, printer-driver choices, and actual loaded paper. A wrong tray setting can look like a hardware failure. [1,8,19]

Sharing, servers & the print queue

VISUAL: Draw digital job envelopes waiting in a neat neon line between laptops, a server, and a printer.



A queue keeps the jobs in line!

QUEUE!

- **Queue.** A waiting line of print jobs. Each job has an owner, status, and options. Paused or failed work can hold up other jobs.
- **Spooler.** Software that stores and schedules print data. In Windows, Print Spooler is a service. A service is a background program that does a system job.
- **Printer share.** A host computer makes its printer available to others. That host and its sharing service must remain available. Permissions control who may print or manage jobs.
- **Print server.** A system that manages shared printer queues. It can centralize access and settings. It may serve network-attached printers as well as locally attached ones.
- **Direct printing.** Each client sends to a network printer directly. A separate PC print server is not always required. Support and management needs decide the design.

STUCK LINE: Confirm the affected queue before canceling work. Restarting a shared spooler can interrupt other users. Test a small new job after recovery. [16]

Secure printing & useful protocols

VISUAL: Draw a purple shield guarding a job until the owner taps a badge in the bright printer lobby.



Hold the job until its owner arrives!

LOCK!

- **Authentication.** Checks who the user is. A badge, account sign-in, or PIN (Personal Identification Number) may identify or unlock a print job.

- **Secure release.** Holds a job until a user releases it at the printer. This reduces unattended private pages. A PIN-release feature does not by itself prove network encryption. [15]

- **Audit log.** A record of events such as who released a job and when. Protect logs and stored jobs according to workplace policy.

- **IPP - Internet Printing Protocol.** A network protocol for submitting and managing jobs. Its default port is TCP 631. TCP means Transmission Control Protocol. Use supported secure configuration for encrypted transport. [10]

- **Legacy printing.** Some printers support raw TCP 9100 or LPD, Line Printer Daemon, on TCP 515. Availability varies. Avoid enabling unneeded services or exposing printers directly to the public internet. [10]

SECURITY HABIT: Change default admin credentials, update firmware, limit access, and use approved secure print and scan settings. Printer memory or storage can hold private documents.

ADF, flatbed & scanning words

VISUAL: Draw a blue-covered book face down on glass, a loose-sheet feeder, and glowing digital pages in neon cyan.



Originals go in. Digital images come out!

SCAN!

- **ADF - Automatic Document Feeder.** Feeds a stack of suitable loose originals into a scanner. Remove staples and clips first. Load face up or down as the device icon and manual show.

- **Flatbed.** A glass scanning surface for one original, a photo, or a supported book. Put the side to scan against the glass. Avoid forcing thick books flat or crushing the lid.

- **Two-sided scanning.** Some feeders flip each sheet; others scan both sides in one pass. An automatic print duplexer does not guarantee a two-sided scanner.

- **OCR - Optical Character Recognition.** Software that tries to turn text in a scanned image into searchable or editable words. It can make mistakes. Check important numbers and names.

- **PDF / JPEG.** Portable Document Format can hold multiple pages. JPEG means Joint Photographic Experts Group and commonly names a compressed image format. Choose what the recipient needs.

LINE CLUE: A line only on feeder scans, with clean flatbed scans and direct prints, points first toward the ADF scan path or narrow glass strip. [17]

Scan destinations: send it safely

VISUAL: Draw glowing scanned pages traveling toward a yellow folder, an envelope, and a smiling cloud.



Give every scan the right address!

SEND!

- **Scan to email.** The printer may use SMTP, Simple Mail Transfer Protocol, to send mail. Configure the approved server, port, security, and authentication. Verify the recipient and attachment limit.
- **Scan to SMB.** Server Message Block is a network file-sharing protocol. The printer needs the correct folder path, credentials, and permission to write a file. Use supported secure settings. [23]

- **Scan to cloud.** An authorized app or service uploads scans to a chosen account and folder. Check network access, account sign-in, and workplace rules for private documents.
- **EWS - Embedded Web Server.** A management website built into many printers. Open the correct printer address using approved secure access. An EWS is not the same thing as an external print server.
- **Verify delivery.** Send a harmless test page. Confirm it reaches the intended destination and opens correctly. A successful local scan does not prove email or folder delivery works. [23]

ACCESS CLUE: If scanning works but saving to a folder fails, check the network path and write permissions before changing the printhead or toner.

Troubleshooting: split the problem

VISUAL: Draw a printer detective comparing a built-in test page, a computer print, and a scan at a colorful desk.



Find the broken step before swapping parts!

THINK!

- **1. Ask and observe.** Read the exact error. Ask when it began, what changed, and who is affected. Check power, display status, supplies, paper, and open covers.
- **2. Test the engine.** Print an internal report or test page from the printer controls. A defect there points toward the device, supplies, settings, or media rather than the computer app.

- **3. Test the path.** If the internal page is clean, try a small known-good file from the computer. Compare another app, user, or client where allowed. Narrow the failure before making changes.
- **4. Separate print and scan.** Compare a direct print, flatbed copy, and ADF copy. Only one bad path helps identify the section to inspect. [17]
- **5. Fix and verify.** Apply one suitable fix, retest the original task, and document what changed. Escalate unsafe or internal faults beyond your training.

DO NOT GUESS FROM ONE SYMPTOM: A faded page might involve toner, ink, settings, media, or another part. Use test results and the model's troubleshooting chart. [14,16]

Print defects: read the clues

●●● **VISUAL:** Draw a careful tech comparing clean, faded, streaked, and doubled print samples under bright magnifiers.



Every mark is a clue, not a verdict!

LOOK!

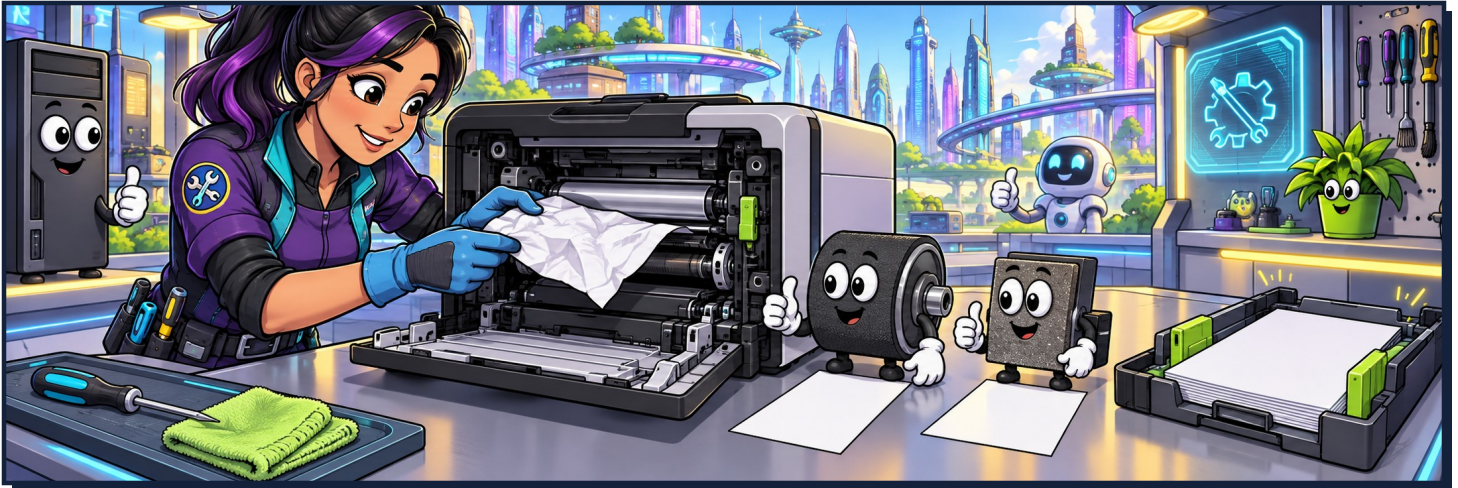
Symptom	Useful first checks
Faded output	Supply level, economy mode, correct media; inkjet nozzle check
Lines or specks	Internal test page, clean path, cartridge/drum or nozzle clues
Repeated / echo image	Model defect chart, repeating interval, drum or fuser path
Loose laser toner	Paper type/settings, approved cleaning, fusing system
Garbled characters	Correct driver and language, app/file, connection

- **Repeating interval.** A damaged rotating part can repeat a mark at a regular spacing. Match the spacing to the model’s service chart. Do not memorize one universal distance for all printers. [14]
- **Color off target.** Check supplies and media settings. Run the approved alignment or calibration routine. A monitor and printed page can look different.

SAVE THE SAMPLE: Keep a test page so you can compare before and after. Replace parts only after the checks support that action. [4,9,14]

Jams, misfeeds & finishing faults

VISUAL: Draw the rescue tech gently clearing an intact jam while roller and separator characters guard the paper line.



Stop, inspect, and pull gently as directed!

STOP!

- **Paper jam.** Follow the display and manual for the correct access door and removal direction. Use the required shutdown and cool-down steps. Pull gently, avoid tearing, and check for scraps.

- **Paper not feeding.** Check the stack, guides, tray seating, supported size, and pickup roller. Do not overload the tray or force thick stock into a path that cannot handle it.

- **Multipage misfeed.** Check damp or stuck sheets and the stack condition. Inspect the separation pad or roller. Several sheets feeding together is different from several copies requested in software.

- **Grinding noise.** Stop the job. Check for obstructions and improperly seated parts when safe. Do not keep running the gears or reach into a moving printer.

- **Staples / hole punch.** Use the approved staple unit and paper limits. Clear a staple jam or empty the punch-waste bin as directed. Keep fingers away from sharp mechanisms. [20]

TRAY NOT RECOGNIZED? Check physical seating, accessory connections, and driver-installed options. Confirm the tray and paper type are supported. [1,8]

Offline, wrong pages & frozen jobs

VISUAL: Draw a laptop and printer checking a paused neon job queue while a technician follows the cable path.



A powered-on printer can still be unreachable!

CHECK!

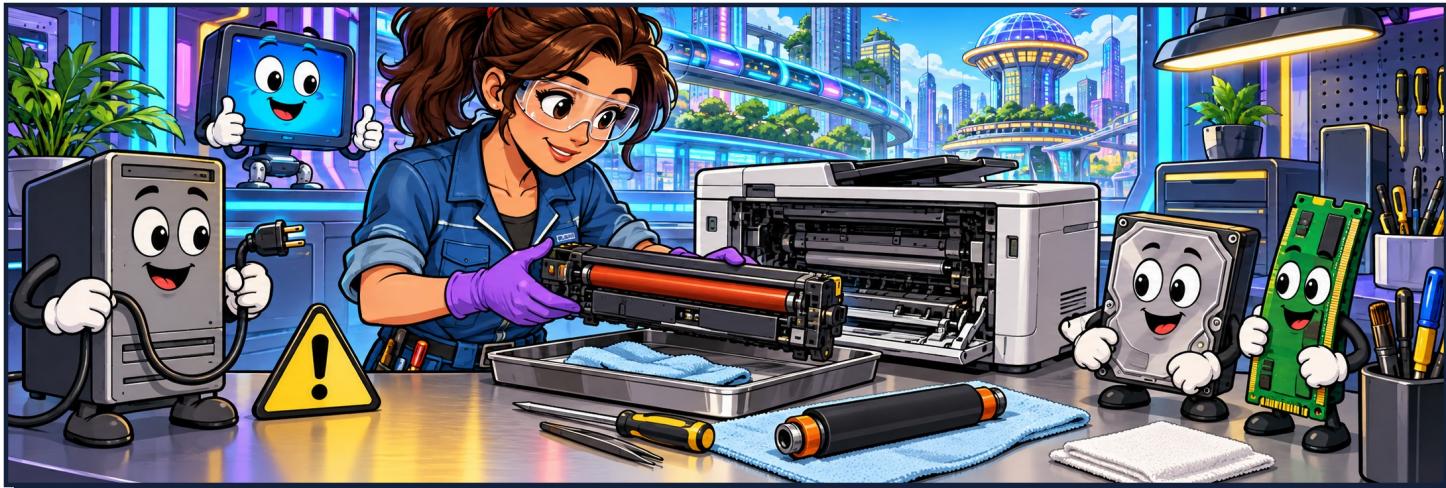
- **Offline / connectivity.** Confirm the intended device, power, cable or Wi-Fi, and current address. Check for a paused queue or offline setting. A printer can be on while its network service is unreachable.
- **Frozen queue.** Inspect the first failing job and its error. Cancel the affected job when appropriate. Restart the spooler only with awareness of other users and queued work. [16]

- **Pending jobs.** The printer may be waiting for the right paper, a closed cover, an error to clear, or secure release. Read status before deleting all jobs.
- **Wrong orientation.** Compare app layout, driver orientation, scaling, and duplex binding. Preview and print one test sheet before repeating a large job.
- **After recovery.** Send a small new job. Confirm it leaves the queue and appears at the right printer. Then retry the original work. Document any changed address or driver.

SCOPE CLUE: One app fails? Inspect that app or file. One PC fails? Inspect its path and driver. Everyone fails? Inspect the shared queue, printer, and network.

Safety & a repeatable care plan

VISUAL: Draw the tech logging service beside a cooled printer, approved cloths, and a clear disconnected plug.



Careful hands keep the whole crew working!

SAFE!

- **Heat and electricity.** Allow hot parts to cool for the time the manual specifies. Do not open power-supply or laser enclosures. Stop if there is smoke, burning odor, or damaged wiring. [12]

- **Toner spills.** Avoid breathing or spreading dust. Do not use compressed air or an ordinary household vacuum. Follow the maker's spill method; approved toner vacuums need suitable fine-particle control. [3,12]

- **SDS - Safety Data Sheet.** Explains a supply's hazards, handling, spill response, and disposal. Use the sheet for the exact product. Follow local recycling and workplace procedures.

- **Cleaning choices.** Use approved tools and fluids for the specific surface. Apply cleaner to a cloth when instructed, not into the machine. Let parts dry as directed before use.

- **Record and repeat.** Log symptoms, page count, supplies, replaced parts, cleaning, and the final test. Schedule maintenance by the manual and actual use.

POWER RULE: Internal hand service and powered cleaning routines have different instructions. A cleaning cycle needs power; handling a hot internal part requires the proper shutdown and cooling steps.

Flashcard mission: say, flip, repeat

VISUAL: Draw the printer and toner buddies studying blank bright cards under gold stars in their neon workshop.



Say it first. Reveal it. Try it again!

RECALL!

- **PowerPoint mode.** Open the companion deck in Slide Show mode. Pick a card. Say the whole answer before pressing REVEAL ANSWER. Use TRY AGAIN, NEXT CARD, or CARD MENU.

- **A short round.** Practice six cards at a time. Keep your own tally. Put missed topics on a short list and review the matching PDF page.

- **Repeat with a gap.** Retry difficult cards later today, tomorrow, and on later days. Also explain why the answer fits a real printer problem.

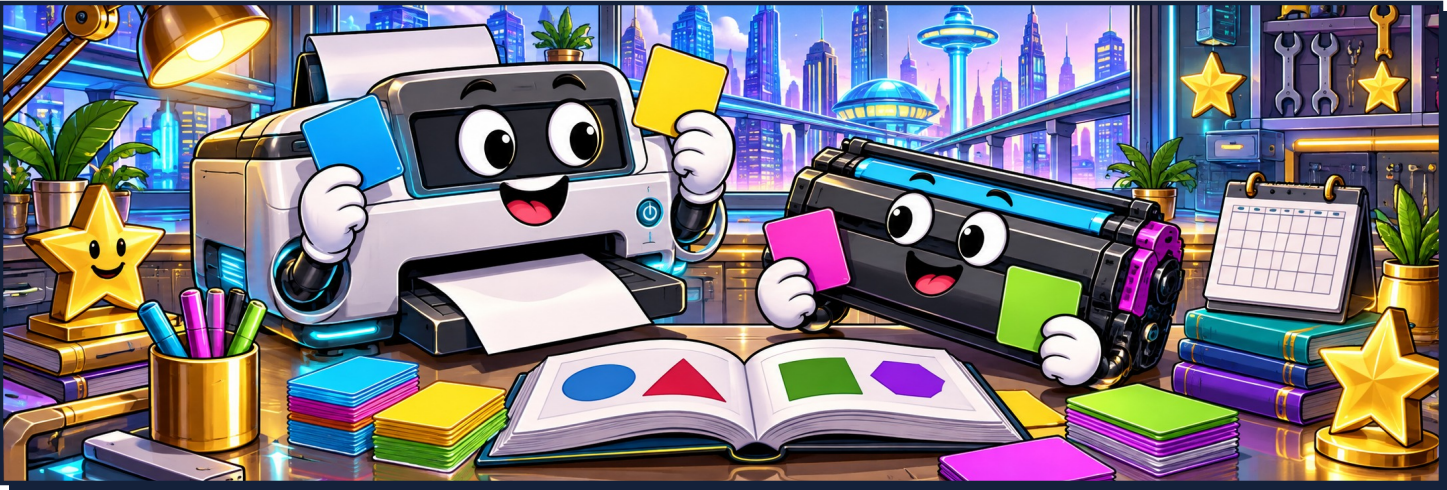
- **Print option.** The next six pages show the same 24 questions and answers. Cover the answer column, speak your answer, then uncover it.

- **What the deck does.** Buttons jump between question and answer slides. No macros are required. It does not save scores or automatically schedule reviews.

MEMORY + REASONING: Learn the laser order by heart. Then explain what each stage does. Rote recall becomes more useful when you can apply it to a symptom.

Flashcards 01-04: cover & recall

●●● **VISUAL:** Draw the printer study team with cyan, yellow, and purple cards and bright gold stars.



Cover the answer. Speak. Check. Repeat!

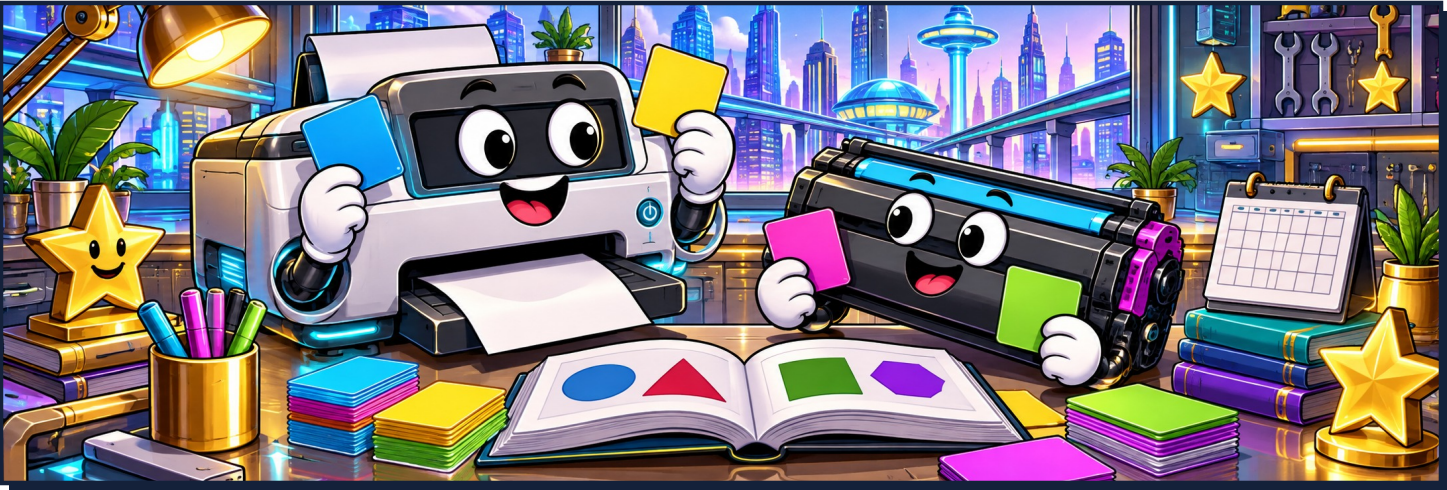
FLIP!

Card / question	Answer to recall
01. Expand MFP and name its common jobs.	Multifunction Printer. It prints, scans, and copies. Some models also fax.
02. Expand PPM and DPI. Which measures speed?	Pages Per Minute measures speed. Dots Per Inch describes dot density.
03. What marking supply does a laser printer use?	Toner powder. A fuser bonds it to paper with heat and pressure.
04. Say all seven laser stages in order.	Processing, Charging, Exposing, Developing, Transferring, Fusing, Cleaning.

Use the matching PowerPoint for clickable answer reveals. For print practice, cover the right column. Review the lesson behind any missed answer.

Flashcards 05-08: cover & recall

VISUAL: Draw the printer study team with cyan, yellow, and purple cards and bright gold stars.



Cover the answer. Speak. Check. Repeat!

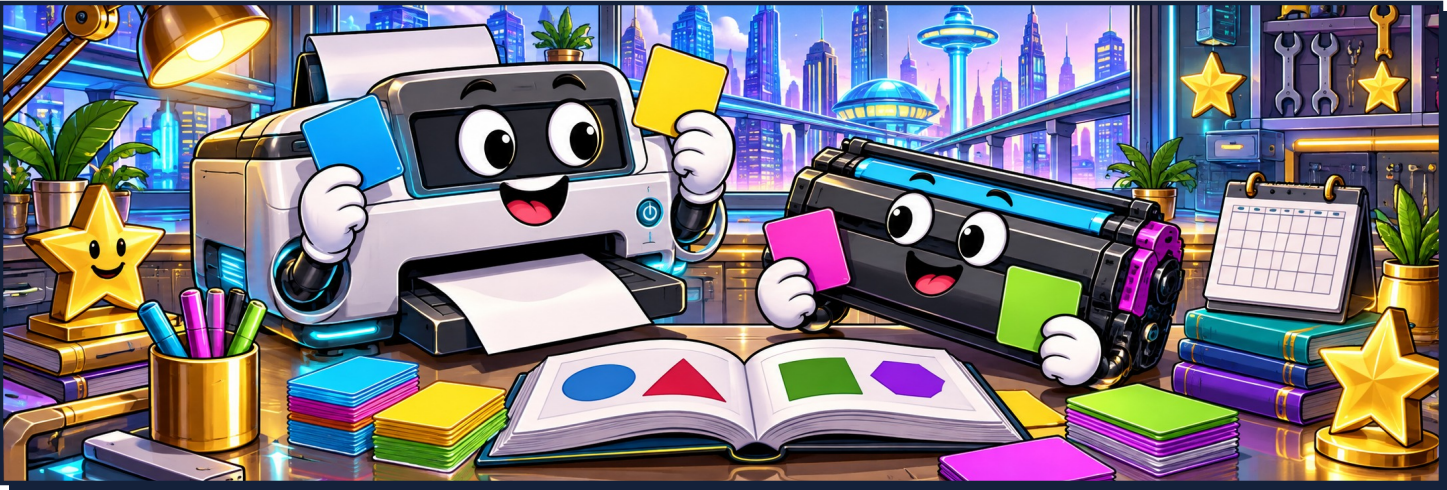
FLIP!

Card / question	Answer to recall
05. What does laser processing prepare?	A page image the print engine can produce, based on the job data.
06. What does charging do? What does exposing do?	Charging gives the drum an even charge. Exposing writes a hidden charge pattern with light.
07. What happens during developing?	Charged toner forms the visible image on the drum.
08. What happens during transferring?	The toner image moves toward the paper, sometimes through an intermediate belt.

Use the matching PowerPoint for clickable answer reveals. For print practice, cover the right column. Review the lesson behind any missed answer.

Flashcards 09-12: cover & recall

●●● **VISUAL:** Draw the printer study team with cyan, yellow, and purple cards and bright gold stars.



Cover the answer. Speak. Check. Repeat!

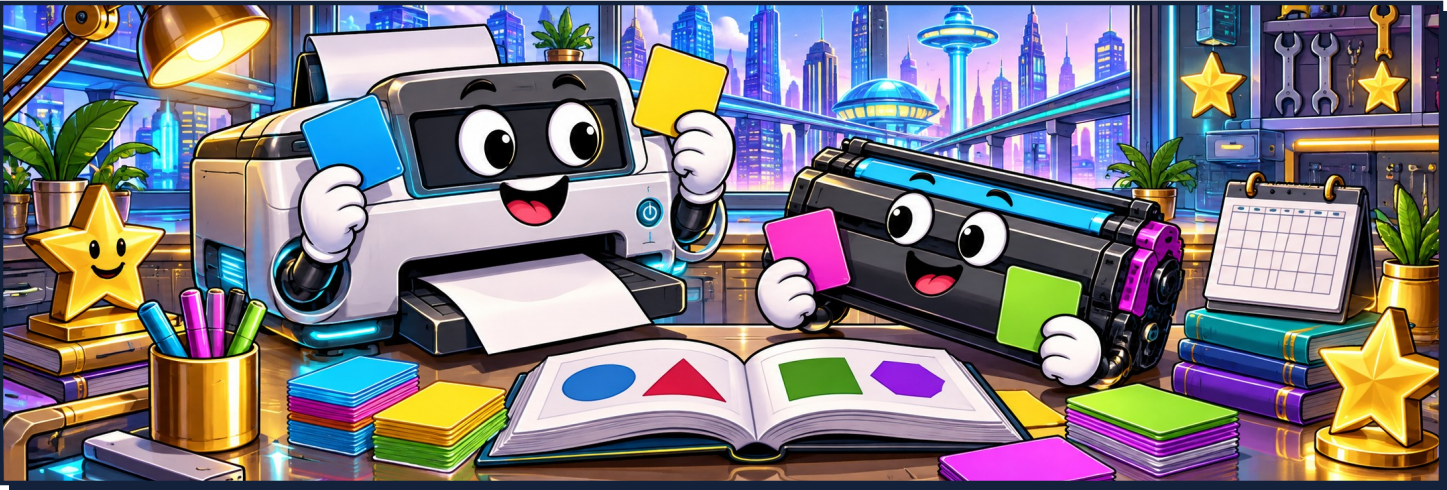
FLIP!

Card / question	Answer to recall
09. What two forces bond toner to paper?	Heat and pressure from the fuser.
10. What does laser cleaning remove?	Leftover toner from the drum so the drum can be prepared for another image.
11. Does every laser maintenance kit contain the same parts?	No. Contents and service intervals depend on the model.
12. What can gaps in a nozzle-check pattern suggest?	Blocked nozzles or an ink-supply problem. Check ink and clean as directed.

Use the matching PowerPoint for clickable answer reveals. For print practice, cover the right column. Review the lesson behind any missed answer.

Flashcards 13-16: cover & recall

●●● **VISUAL:** Draw the printer study team with cyan, yellow, and purple cards and bright gold stars.



Cover the answer. Speak. Check. Repeat!

FLIP!

Card / question	Answer to recall
13. What is alignment meant to adjust?	Where the ink lands, helping correct offset or doubled lines.
14. Which thermal method needs special paper? Which uses ribbon?	Direct thermal uses heat-sensitive paper. Thermal transfer uses a ribbon and suitable media.
15. Which printer type strikes a ribbon to make multipart copies?	Impact, such as dot matrix.
16. Which part grabs a sheet? Which holds back extras?	The pickup roller grabs a sheet. A separation pad or roller holds back extra sheets.

Use the matching PowerPoint for clickable answer reveals. For print practice, cover the right column. Review the lesson behind any missed answer.

Flashcards 17-20: cover & recall

VISUAL: Draw the printer study team with cyan, yellow, and purple cards and bright gold stars.



Cover the answer. Speak. Check. Repeat!

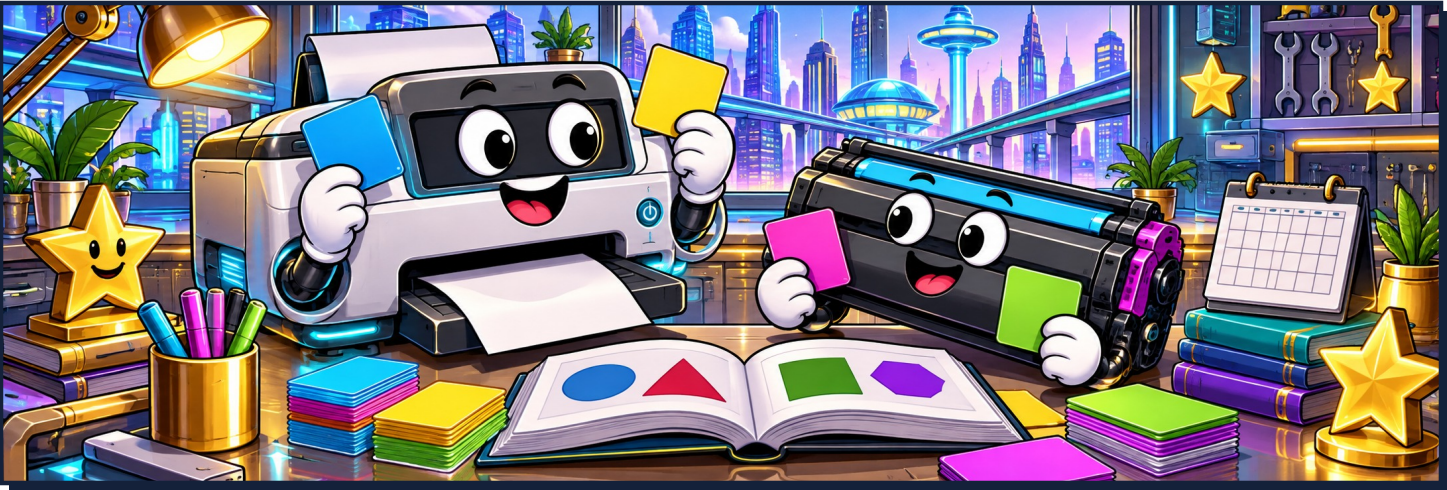
FLIP!

Card / question	Answer to recall
17. Expand ADF. How is its job different from a duplexer?	Automatic Document Feeder feeds scan originals. A print duplexer routes paper for two-sided printing.
18. What is PCL? What does PS mean in a printer driver?	PCL is Printer Command Language. PS means PostScript.
19. Where do a driver and printer firmware do their jobs?	The driver helps the computer send print data. Firmware runs inside the printer.
20. What is the print spooler's job?	It stores and schedules print data for queued jobs.

Use the matching PowerPoint for clickable answer reveals. For print practice, cover the right column. Review the lesson behind any missed answer.

Flashcards 21-24: cover & recall

VISUAL: Draw the printer study team with cyan, yellow, and purple cards and bright gold stars.



Cover the answer. Speak. Check. Repeat!

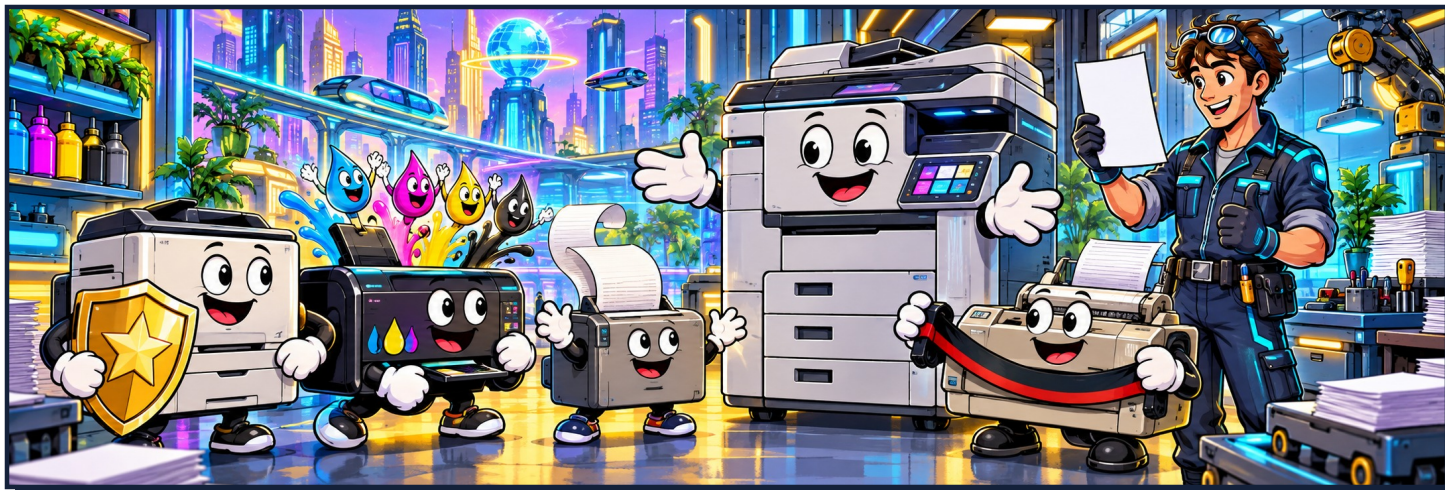
FLIP!

Card / question	Answer to recall
21. Why hold a job until a PIN or badge releases it?	To reduce private pages sitting unattended before the owner arrives.
22. ADF scans have a line. Flatbed scans and direct prints are clean. Check what first?	The ADF scan path, especially the narrow scan-glass strip.
23. Expand SMB and OCR and explain their jobs.	Server Message Block shares files. Optical Character Recognition turns image text into searchable or editable words.
24. Laser toner rubs off a page. What should you check?	Paper type and media settings, then the cleaning and fusing guidance for that model.

Use the matching PowerPoint for clickable answer reveals. For print practice, cover the right column. Review the lesson behind any missed answer.

Sources: technology & setup

VISUAL: Draw the printer crew in a bright neon library holding blank reference books.



Check the facts. Then practice the clues!

READ!

1. CompTIA: Core 1 exam objectives

[Open primary reference](#)

2. HP: Laser printing mechanics

[Open primary reference](#)

3. HP: Laser maintenance kits

[Open primary reference](#)

4. Epson: Faint prints and nozzle gaps

[Open primary reference](#)

5. Zebra: Direct thermal versus transfer

[Open primary reference](#)

6. Zebra: Printer cleaning

[Open primary reference](#)

7. Epson: Multipart forms

[Open primary reference](#)

8. HP: Selecting a printer driver

[Open primary reference](#)

9. Epson: Printhead alignment

[Open primary reference](#)

10. PWG: Internet Printing Protocol guide

[Open primary reference](#)

11. Apple: About AirPrint

[Open primary reference](#)

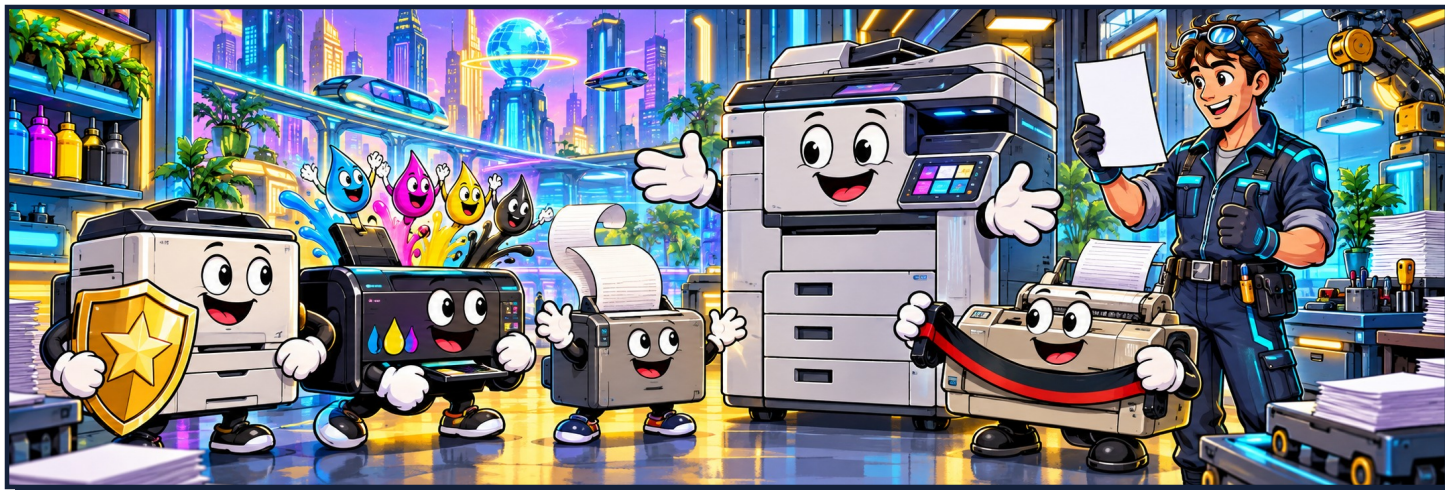
12. HP: Printer safety information

[Open primary reference](#)

Reviewed September 19, 2026. Sources include model-specific examples: use your own printer's manual for exact service steps. Independent study material; no CompTIA or Per Scholas endorsement.

Sources: care & troubleshooting

VISUAL: Draw the printer crew in a bright neon library holding blank reference books.



Check the facts. Then practice the clues!

READ!

13. HP: LaserJet cleaning procedures

[Open primary reference](#)

14. HP: Laser print-quality troubleshooting

[Open primary reference](#)

15. HP: Private and stored print jobs

[Open primary reference](#)

16. Microsoft: Offline printers and queues

[Open primary reference](#)

17. HP: Lines only in ADF scans

[Open primary reference](#)

18. HP: Workload and speed specification example

[Open primary reference](#)

19. HP: Standard paper sizes

[Open primary reference](#)

20. HP: Stapler and hole-punch jams

[Open primary reference](#)

21. HP: Unsupported PCL language error

[Open primary reference](#)

22. HP: Firmware update precautions

[Open primary reference](#)

23. HP: Network scan setup

[Open primary reference](#)

24. Epson: Inkjet drop technology

[Open primary reference](#)

Reviewed September 19, 2026. Sources include model-specific examples: use your own printer's manual for exact service steps. Independent study material; no CompTIA or Per Scholas endorsement.

STAR QUIZ TIME

 **VISUAL:** Draw the printer and toner buddies celebrating under three bright gold stars in the neon study room.



Three missions. You have the tools!

YOU GOT IT!

1. A laser page looks right, but toner rubs off. Which stage bonds toner to the page, and what should you check before replacing a part?
2. A receipt printer uses heat-sensitive paper and no ribbon. Which printer method is this?
3. Direct prints and flatbed scans look clean. ADF scans have a dark line. What does ADF mean, and where should you look first?

ANSWERS - CHECK YOUR HERO SKILLS!

1. **Fusing.** Heat and pressure bond toner. Check the paper type and media settings, then follow the model's cleaning and fuser guidance. Great work using the clue instead of guessing!
2. **Direct thermal.** Heat changes the coated paper. Thermal transfer uses a ribbon. Nice job spotting the supply difference!
3. **Automatic Document Feeder.** Inspect and clean the ADF scan path, especially the narrow glass strip, as directed. You separated the scan problem from the print engine!